MIL-R-49353(ER)
30 January 1979

MILITARY SPECIFICATION

RF CONVERTER CV-3168( )/ULR-17(V)

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers the requirements for the construction and testing of the RF Converter, CV-3168( )/ULR-17(V). The RF Converter unit will frequency convert RF signals in the range from 1 to 30 MHz to an IF signal centered at 455 kHz. This RF Converter unit is to be used in conjunction with Receiver, Radio R-1849( )/ULR-17(V), Amplifier, Intermediate Frequency AM-6608( )/ULR-17(V), Interconnecting Box J-3176( )/ULR-17(V) and Interconnecting Box J-3177( )/ULR-17(V) to complete a matched channel receiving subsystem. (see Figure 1).

2. APPLICABLE DOCUMENTS.

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or requests for proposal, form a part of the specification to the extent specified herein.

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, ERADCOM, Fort Monmouth, N.J. 07703, ATTN: DELET-DP, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5820

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SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-Handling, Wood
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade), and Cut Shapes
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Triple-Wall
PPP-S-760	Strapping, Nonmetallic (and Connectors)

MILITARY

MIL-P-116	Preservation, Method of
MIL-P-11268	Parts, Materials, and Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish For Ground Signal Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, For Packaging

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-147	Palletized and Containerized Unit Loads 40 inch x 48 inch Pallet
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment

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SPECIFICATIONS, STANDARDS (MILITARY CONT'D)

MIL-STD-461A	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics and Measurement of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Equipment

DRAWINGS

SM-A-689134	Acceptance Test Procedure for RF Converter
SM-D-2014502	CV-3168( )/ULR-17(V) RF Converter
DLSM-B-783853	CV-3168( )/ULR-17(V) Data List

(Copies of specifications, standards, and drawings required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS.

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and the drawings listed in the USAERADCOM Data List DLSM-B-783853. The RF Converter, CV-3168( )/ULR-17(V) top assembly drawing number is SM-D-2014502.

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3.3 Parts, materials and process; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein shall apply.

3.3.1 Finish. The CV-3168 shall be finished in accordance with MIL-F-14072 and the drawings listed on DLSM-B-783853.

3.4 Marking.

3.4.1 General. Marking shall conform to MIL-M-13231 and MIL-STD-130 (see 4.4). Front panel marking shall be Group I as described in MIL-M-13231.

3.4.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.3 Serial numbers. Each CV-3168 shall have a serial number.

3.5 Performance.

3.5.1 General characteristics. The RF Converter, CV-3168, converts RF input frequencies in the range 1 MHz to 30 MHz to an IF frequency of 455 kHz. Two or more CV-3168 converters, when used with associated Amplifier,IF units, AM-6608 and HF receiver, R-1849, provide a multichannel phase and gain matched receiving capability.

3.5.2 Frequency range. The RF Converter, CV-3168, shall operate in the frequency range 1.0 MHz to 29.9999 MHz.

3.5.3 Gain. The gain of the RF Converter, CV-3168, shall be as specified in Table I.

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TABLE I. Gain

FREQ (MHz)	*MAX GAIN (db)	*MIN GAIN (db)
1.0 - 1.9999	22.5	20.0
2.0 - 2.9999	22.5	20.0
3.0 - 4.9999	22.5	20.0
5.0 - 8.9999	22.5	20.0
9.0 -14.9999	22.5	20.0
15.0 -21.9999	22.0	19.5
22.0 -29.9999	21.5	19.0

*NOTE: With all three local oscillator inputs at a level of 0 ± 0.25 dbm and all attenuators at minimum loss, the gain of the CV-3168 shall be within the following specified maximum and minimum values over the specified frequency ranges.

3.5.4 Noise figure. The noise figure of the RF Converter, CV-3168, shall not exceed 14db.

3.5.5 Inputs.

3.5.5.1 Antenna input. The antenna input shall have a nominal impedance of 50 ohms. The VSWR shall be no greater than 2.5 to 1.

3.5.5.2 Local oscillator inputs. The RF Converter, CV-3168, shall have three local oscillator (LO) inputs (112.0450 to 141.5449 MHz, 101 MHz, 11 MHz). Input impedance shall be 50 ohms (nominal) with a VSWR no greater than 2.0 to 1. For a constant RF input, the 455 kHz IF output shall not vary more than ± 1db for local oscillator inputs of -2.5dbm to +4dbm (reference level shall be the IF output with all LO inputs at 0dbm).

3.5.5.3 Attenuator control inputs. There shall be four, low power TTL compatible, attenuator control inputs. A voltage of 2.4 to 5.0 volts on any of the attenuator input lines shall activate the corresponding attenuator (3.5.6) reducing the gain by 10db. An input voltage of 0 to 0.8 volts on any of the attenuator lines shall deactivate the corresponding attenuator.

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3.5.5.4 Band select inputs. There shall be seven, low power TTL compatible, band select inputs. A voltage of 2.4 to 5.0 volts on a band select input line shall insert the corresponding preselector filter in the RF signal path. A voltage of 0 to 0.8 volts shall remove the corresponding preselector filter from the RF signal path.

3.5.6 Attenuator accuracy. There shall be four RF input attenuators. Each attenuator shall be controlled by one of the four attenuator control inputs. When activated, each attenuator shall reduce the gain of the RF Converter, CV-3168, by 10 $\pm$ 0.4db.

3.5.7 Power.

3.5.7.1 Input power. The input voltage to the RF Converter, CV-3168, shall be 12 $\pm$ 0.2Vdc. Maximum input current shall be 0.60 amps while supplying the voltages and currents specified in 3.5.7.2.

3.5.7.2 Output voltages. The RF Converter shall meet the requirements of 3.5.1 through 3.5.7.1 while supplying the following outputs:

+5 $\pm$ 0.1Vdc @ 23ma

+2.5 $\pm$ 0.025Vdc @ 26ma

-12 $\pm$ 0.24Vdc @ 13ma

3.6 Environmental. The CV-3168 hereafter called equipment shall meet the following service conditions without degradation in specified performance of 3.5 unless specified herein.

3.6.1 Temperature-altitude. The equipment shall be operable without degradation in specified performance at altitudes up to 30,000 feet above sea level and shall withstand air transportation at altitudes up to 50,000 feet above sea level. The equipment shall be capable of continuous operation without degradation in specified performance at ambient temperatures as low as -40 degrees F and as high as +131 degrees F, with intermittent operation to +160 degrees F. The equipment shall withstand exposure to ambient air temperature as low as -70 degrees F and as high as +160 degrees F. Testing shall be in conformance with the test requirements of 4.9.1.

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3.6.2 Humidity. The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to relative humidities as high as 100 percent at all ambient air temperatures to +85 degrees F and relative humidities corresponding to a dew point of +85 degrees F at all temperatures between +85 degrees F and +120 degrees F, and low relative humidity of 5 percent at +120 degrees F. (see 4.9.2).

3.6.3 Sand and dust. The equipment shall withstand, in both operating and nonoperating conditions, exposure to fine sand and dust particles with wind speeds of up to 20 knots. After testing as specified in 4.9.3, the equipment shall meet the specified performance requirements.

3.6.4 Salt fog. In both operating and nonoperating conditions the equipment shall withstand the salt fog test of 4.9.4. Upon completion of test, the CV-3168, where examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surfaces; or exfoliation of metal. There shall be no clogging or binding of controls or other moving parts after subjecting the CV-3168 to the 48 hour salt fog exposure period or after subjection to the subsequent 48-hour drying period.

3.6.5 Fungus. After the fungus test of 4.9.5, the equipment shall be visually examined, using a 10-power magnifier. The CV-3168 shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.3.3) as evidenced by growth colonization (see 6.3.2) which includes branching (see 6.3.1) and sporulation (see 6.3.4) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.3.5) shall not be included in this evaluation.

3.6.6 Vibration and shock. The equipment shall withstand vibration and shock induced during field transport, by military vehicle, by rough handling, and mounted in military aircraft. Vibration isolation will be provided by the systems integrator when installed in the airframe.

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3.6.7 Bench handling. The equipment shall withstand bench handling.

3.6.8 Explosive atmosphere. The equipment shall not cause ignition of an ambient explosive gaseous mixture with air when operating in such an atmosphere.

3.7 Electromagnetic interference (EMI). The equipment shall comply with the following emission and susceptibility requirements of Notice 4, MIL-STD-461A.

CE01	CS01	RE02	RS03.2
CE03	CS02	RE02.1	
CE04	CS04		
CE05	CS06		
CE06			

3.8 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.8).

3.9 Burn-in. Each unit on order shall meet the requirements herein, without subsequent processing, after subjection to burn-in as specified in 4.6.

3.10 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268

General requirements for plastic material and plastic parts.
Wiring and cabling, Including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Flux and cleaning agent for soldering
Process for soldering
Cleaning
Riveting
General requirements for securing of parts
Cleaning after welding
Mechanical operation of controls
Process for welding
Self-tapping screws

b. In this specification:

3.3.1 Finish

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3.11 Systems safety engineering.

3.11.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.10).

3.11.2 Edge rounding exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.05 inch (1.3 mm).

3.11.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.12 Human factors engineering. Design configuration as affecting operation, maintainability and handling shall be maintained through compliance of configuration criteria to the requirements of MIL-STD-1472.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging). (see 4.3).

b. Inspection covered by subsidiary documents. (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging. (see 4.5).

(2) Quality conformance inspection of packaging. (see 4.11).

4.3 First article. Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish four First Article units of RF Converter, CV-3168( )/ULR-17(V).

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in subsidiary documents covering the items listed in 4.4 and the inspections specified for Group A, Group B, and Group C (see Table III, IV and V respectively). The inspection shall be performed in the following order: 4.4, Group A and Group B for all units and designated units for specific Group C tests in the order shown in Table II.

TABLE II. Environmental test for first article inspection.

INSPECTION (Note 1)	UNIT 1	UNIT 2	UNIT 3	UNIT 4
Vibration, equipment		1		
Temperature-altitude	2			
Shock, bench handling				1
Humidity	3			
Fungus (Note 2)		2		
Salt fog (Note 2)			1	
Sand and dust (Note 2)				2
Explosive atmosphere			2	
EMI	1			

NOTE 1: Other first article inspection may precede, follow, or be interspersed between the tests.

NOTE 2: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification.

ITEM	WHERE REQUIRED
Finish Protective	3.3.1
Marking	3.4
Parts, Materials and Processes	3.3

4.5 Quality conformance inspection of equipment before packaging. Each unit of each lot of equipment shall be burned-in prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the general inspection levels, and AQLs indicated in Table III.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and to the special inspection level of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans for the AQL's listed in Table IV shall be S-4. All defects shall be considered major.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order satisfactory to the Government.

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TABLE III. Group A inspection.

INSPECTION	REQ. PARA.	INSP. PARA.	AQL MAJOR	AQL MINOR
Visual and mechanical	3.10	4.10	1.0%	4.0%
Electrical				
Frequency range	3.5.2	4.7.1	1.0%	*
Gain	3.5.3	4.7.2	1.0%	*
Noise figure	3.5.4	4.7.3	1.0%	*
Attenuator control inputs	3.5.5.3	4.7.6	1.0%	*
Band select inputs	3.5.5.4	4.7.7	1.0%	*
Attenuator accuracy	3.5.6	4.7.8	1.0%	*
Power	3.5.7	4.7.9	1.0%	*

*All electrical defects shall be considered major.

TABLE IV. Group B inspection

INSPECTION	REQ PARA.	TEST PARA.	AQL
Antenna input	3.5.5.1	4.7.4	1.0%
Local oscillator inputs	3.5.5.2	4.7.5	1.0%
Interchangeability	3.8	4.8	6.5%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1. The inspection shall consist of the inspection specified in Table V.

4.5.3.1 Sampling for Group C inspection. Four samples of the complete equipment shall be selected at random, 1 sample for each Subgroup I through and including Subgroup IV in Table V. The first samples shall be selected at the start of production from the first quality conformance inspection lot. These samples shall constitute the Group C requirements for the first 25 units produced. Thereafter, 4 samples of the complete equipment shall be selected at random, 1 sample for each Subgroup I through Subgroup IV in Table V from every 75 units and fraction thereof produced. Salt fog and fungus shall not be repeated after the first production lot Group C inspection.

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TABLE V. Group C inspection.

INSPECTION	REQ. PARA.	TEST PARA.
Sub-Group I		
Temperature-altitude	3.6.1	4.9.1
Humidity	3.6.2	4.9.2
Sub-Group II		
Vibration	3.6.6	4.9.6
Sub-Group III		
Shock	3.6.7	4.9.7
Sub-Group IV		
EMI	3.7	4.9.9
Sand and dust 1/	3.6.3	4.9.3
Salt fog 1/	3.6.4	4.9.4
Fungus 1/	3.6.5	4.9.5
Explosive atmosphere	3.6.8	4.9.8

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order acceptable to the Government.

4.5.3.3 Group C failure. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

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4.6 Burn-in. Every equipment shall be subjected to a 100 hour burn-in. The last 24 hours of the burn-in shall be failure free. If an equipment fails during the last 24 hours failure free period, it shall be repaired, returned to test, and the test shall continue until a 24 hour failure free period has been achieved.

4.6.1 Compliance. Units passing burn-in shall be submitted to the electrical measurements of Group A and B inspection without benefit of repair, cleaning, realignment or other changes, unless otherwise specified herein. If after successful completion of burn-in, the equipment fails either of the electrical measurements in Group A or Group B inspection, the equipment shall be reworked and resubmitted to the last 24 hours of the burn-in period and shall successfully complete the 24 hour failure free period of burn-in prior to being resubjected to Groups A and B inspection.

4.6.2 Conditions preceding burn-in. A thermal survey and vibration survey shall be conducted in accordance with MIL-STD-781. The results of the required time to stabilize at the low temperature shall determine the length of the temperature cycle to be conducted during burn-in.

4.6.3 Test conditions. Each equipment shall be subjected to the following test conditions during burn-in:

Temperature -30°C to 55°C (-22°F to 131°F).

Temperature cycling - Temperature cycle shall be: time to stabilize at low temperature as determined by the thermal survey, time to stabilize at high temperature, six (6) hours at the high temperature.

Vibration - 2.2G ± 10% peak acceleration value at any non-resonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of operating time.

Equipment ON-OFF cycling - Equipment OFF during the cooling cycle and ON during the heating cycle.

Input voltage - Nominal voltage of 110 volts +5%, -2%

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4.6.3 Test conditions (cont'd).

Input voltage cycling - The input voltage shall be maintained at 110% nominal for one-third of the equipment "ON" cycle, at the nominal value for the second one-third of the equipment "ON" cycle and 90% for the final one-third of the equipment "ON" cycle. This cycling procedure is to be repeated continuously throughout burn-in.

4.6.4 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this specification and the contractors approved test procedure.

4.7 Performance.

4.7.1 Frequency range. The CV-3168 shall meet the requirements of 3.5.2 when tested in accordance with 5.1 of Test Procedure SM-A-689134.

4.7.2 Gain. The gain of the CV-3168 shall meet the requirements of 3.5.3 when tested in accordance with 5.1 of Test Procedure SM-A-689134.

4.7.3 Noise figure. The CV-3168 shall meet the requirements of 3.5.4 when tested in accordance with 5.2 of Test Procedure SM-A-689134.

4.7.4 Antenna input. The CV-3168 shall meet the requirements of 3.5.5.1 when tested in accordance with 5.3 of Test Procedure SM-A-689134.

4.7.5 Local oscillator inputs. The CV-3168 shall meet the requirements of 3.5.5.2 when tested in accordance with 5.4 of Test Procedure SM-A-689134.

4.7.6 Attenuator control inputs. The CV-3168 shall meet the requirements of 3.5.5.3 when tested in accordance with 5.5 of Test Procedure SM-A-689134.

4.7.7 Band select inputs. Satisfying the requirements of 3.5.3 shall indicate compliance with the requirements of 3.5.5.4.

4.7.8 Attenuator accuracy. The CV-3168 shall meet the requirements of 3.5.6 when tested in accordance with 5.5 of Test Procedure SM-A-689134.

4.7.9 Power. The CV-3168 shall meet the requirements of 3.5.7 when tested in accordance with 5.6 of Test Procedure SM-A-689134.

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4.8 Interchangeability.

4.8.1 Inspection for dimensional interchangeability. The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirement of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, keying pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major units.

e. Size and form of special threads.

4.8.2 Electrical interchangeability. All like modules and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment as determined by the test specified in 3.8 or 4.8.1. However, electrical retuning and realignment of modules during production assembly operations may be expected and are permissible.

4.9 Environmental. The CV-3168 shall be tested in accordance with the methods and procedures of MIL-STD-810 given below. The limits of allowable performance degradation shall be as specified in 3.5 of this specification, and shall form the criteria for passing or failing each environmental test unless stated otherwise. Successful completion of any particular test shall be considered evidence of meeting the corresponding environmental condition.

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4.9.1 Temperature-altitude. The equipment shall be subjected to the test of Method 504 of MIL-STD-810 for Class 4 equipment with the following changes and additions:

a. The temperature for Step 1 shall be -70 degrees F.

b. The temperature for Steps 2 and 3 shall be -40 degrees F.

c. The procedure shall include a one hour exposure (nonoperating) to a pressure of 3.4 inches of Hg simulating air transportation at 50,000 feet at ambient temperature.

Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.5.

4.9.2 Humidity. The equipment shall be subjected to the test of Method 507, Procedure II of MIL-STD-810. In Step 6, measurements shall be taken during the first 5 hours of each cycle. Inability of the CV-3168 to meet the requirements of 3.6.2 shall constitute failure of this test.

4.9.3 Sand and dust. The equipment shall be tested in accordance with Method 510, Procedure 1 of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.5.

4.9.4 Salt fog. The equipment shall be tested in accordance with Method 509, of MIL-STD-810. Operation of the equipment shall not be required, and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the test items surfaces that can be attributed to flaky pitted, blistered, or otherwise loosened finish or metal surface.

4.9.5 Fungus. The equipment shall be subjected to the fungus test specified in Method 508, Procedure I of MIL-STD-810. There shall be abundant growth colonization (6.3.2) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.6.5 shall constitute failure of this test.

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4.9.6 Vibration. The equipment shall be subjected to the vibration test of MIL-STD-810, Method 514, Procedure 1, Curve C, from Figure 514.2-2. Prior to testing, the equipment shall meet the full specification performance for the requirements of 3.5.

4.9.7 Bench handling. The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance as specified in 3.5.

4.9.8 Explosive atmosphere. The equipment shall be subjected to the test of Method 511, Procedure I of MIL-STD-810.

4.9.9 EMI. The equipment shall be tested for compliance with the requirements of 3.7. The test shall be in accordance with Notice 3, MIL-STD-462 as implemented by a contractor prepared, government approved EMI test plan.

4.10 Visual and mechanical inspection. The CV-3168 shall be examined for the defects listed in MIL-STD-252. An inspection shall also be performed to verify compliance with those portions of 3.11 (Systems Safety Engineering) which can be determined visually.

4.11 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of 5 herein. The rough handling test will be performed only when specified (see 6.2).

4.12 Human engineering verification. Maintenance of design configuration to the requirements of 3.12 shall be verified by visual and mechanical inspection.

5. PACKAGING.

5.1 Preservation. Preservation shall be Level A, B or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. RF Converter, CV-3168( )/ULR-17(V) shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. RF Converter CV-3168( )/ULR-17(V) shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods/submethods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed submethod IC-1.

5.1.1.4.2 RF Converter CV-3168( )/ULR-17(V). Each RF Converter CV-3168( )/ULR-17(V) shall be individually unit packed Method III as follows: Cushion on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be as specified in MIL-STD-1188.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

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5.2.1 Level A. A quantity of RF Converter CV-3168( )/ ULR-17(V), unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.2 Level B. A quantity of RF Converter CV-3168( )/ ULR-17(V), unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

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5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of RF Converter,CV-3168( )/ULR-17(V), unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES.

6.1 Intended use. The CV-3168 is one of the high frequency components of Receiver Set, Countermeasures AN/ULR-17(V) which provides a monitoring capability of signals in the high frequency band of 0.5 to 30.0 MHz.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Marking and shipping of equipment.

c. Place of final inspection.

d. Technical literature required.

e. Level A, Level B or Commercial preservation and packing (see Section 5).

f. When rough handling test is required (see 4.11).

6.3 Definitions.

6.3.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.3.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.4 Sporulation. Sporulation is the formulation of minute unicellular reproductive or dormant bodies, called spores.

6.3.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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6.3.6 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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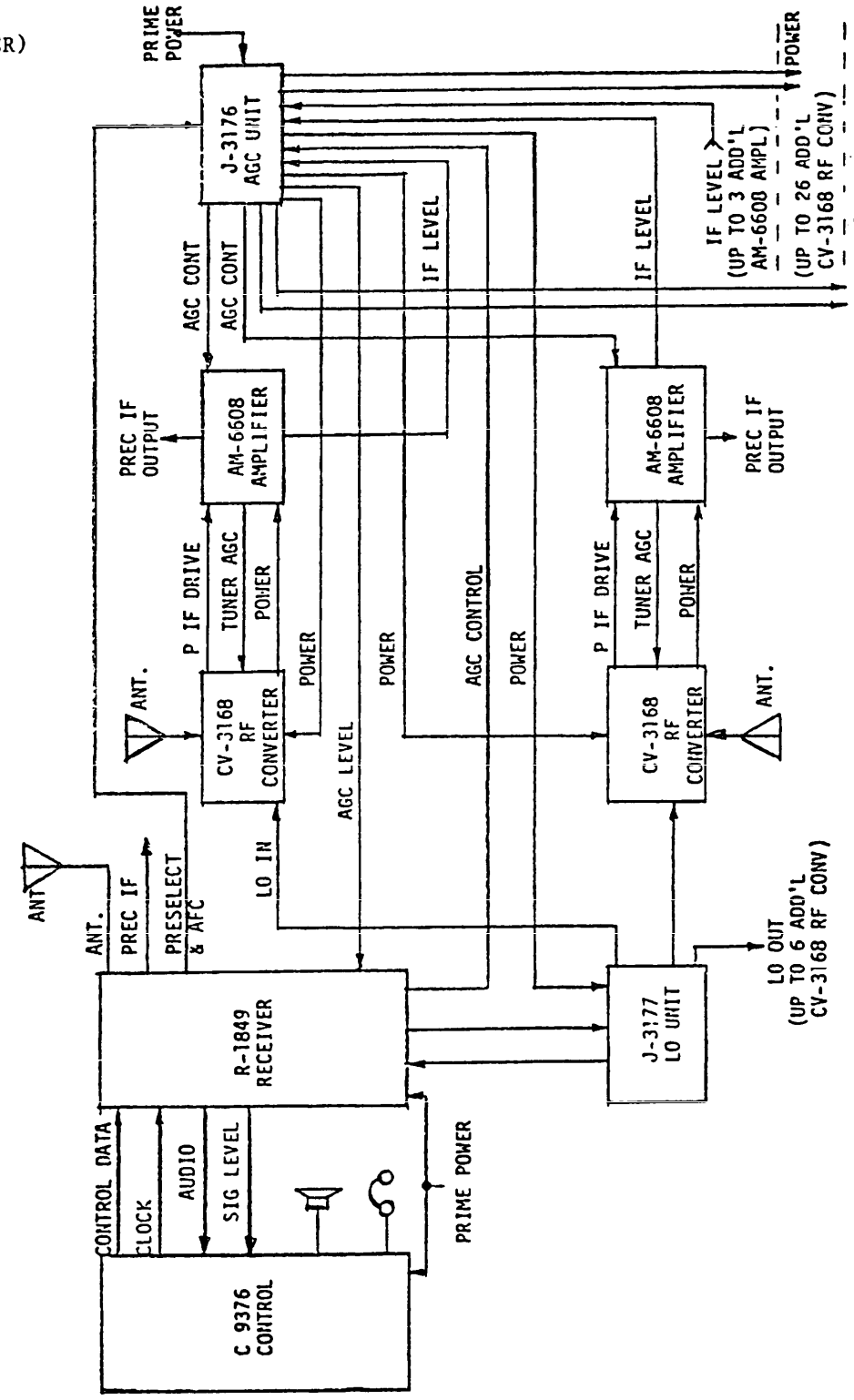


FIGURE 1 - MULTICHANNEL PHASE AND GAIN MATCHED RECEIVING SUBSYSTEM

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DOCUMENT IDENTIFIER AND TITLE: MIL-R-49353(ER) RF CONVERTER CV-3168( )/ULR-17(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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